Predicting Falcon 9 Landing Success: A Data-Driven Approach to Saving Millions.

Capstone Project - IBM Data Science Professional Certificate

https://github.com/andyjoap/Final-Project-SpaceX

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EXECUTIVE SUMMARY

•**Objective:** Develop a machine learning pipeline to predict the success of SpaceX Falcon 9 first-stage landings.

•**Problem:** Each launch costs roughly **$62 million**. Reusing the first stage saves millions, making accurate predictions a high-value asset for mission planning.

Key Findings:

- Payload mass and orbit type significantly influence success.
- Launch sites are strategically located near coastlines for safety.
- **Best Model:** Decision Tree (o el que te haya dado mejor), reaching an accuracy of **[Tu valor, ej. 94% o 83%]**.

INTRODUCTION

- **Background**: SpaceX has revolutionized space travel by introducing reusable rocket technology. The Falcon 9's first stage can land on ground pads or autonomous spaceport drone ships (ASDS).

- **The Problem**: Landing failure results in the loss of expensive hardware. Predicting these outcomes helps in risk assessment and insurance cost estimation.

- **Objective**: To analyze historical launch data and build a classification model that predicts whether the first stage will land successfully based on mission parameters.

Data Collection & Wrangling Methodology

Data Collection: SpaceX REST API and Web Scraping (Wikipedia) for historical launch data.
Data Wrangling: Handling missing values in PayloadMass, creating dummy variables for categorical features (Orbit, Site), and label encoding landing outcomes (1 = Success, 0 = Failure).
Analysis Tools: Python (Pandas, Numpy), SQL for structured queries, Folium for spatial analysis, and Scikit-Learn for modeling.

EDA & Interactive Visual Analytics Methodology

- Exploratory Data Analysis (EDA): Used Pandas and Seaborn to find correlations between variables like Payload Mass and Launch Site.
- SQL Analysis: Leveraged SQLite to perform complex queries, such as identifying the total number of successful missions and ranking launch sites by volume.
- Exploratory Data Analysis (EDA): Performed comprehensive data exploration using Pandas and Seaborn to identify correlations between payload mass, launch sites, and mission success.
- Statistical Visualization: Generated various charts (Scatter plots, Bar charts, and Line graphs) to visualize trends over time and the performance of different rocket versions.

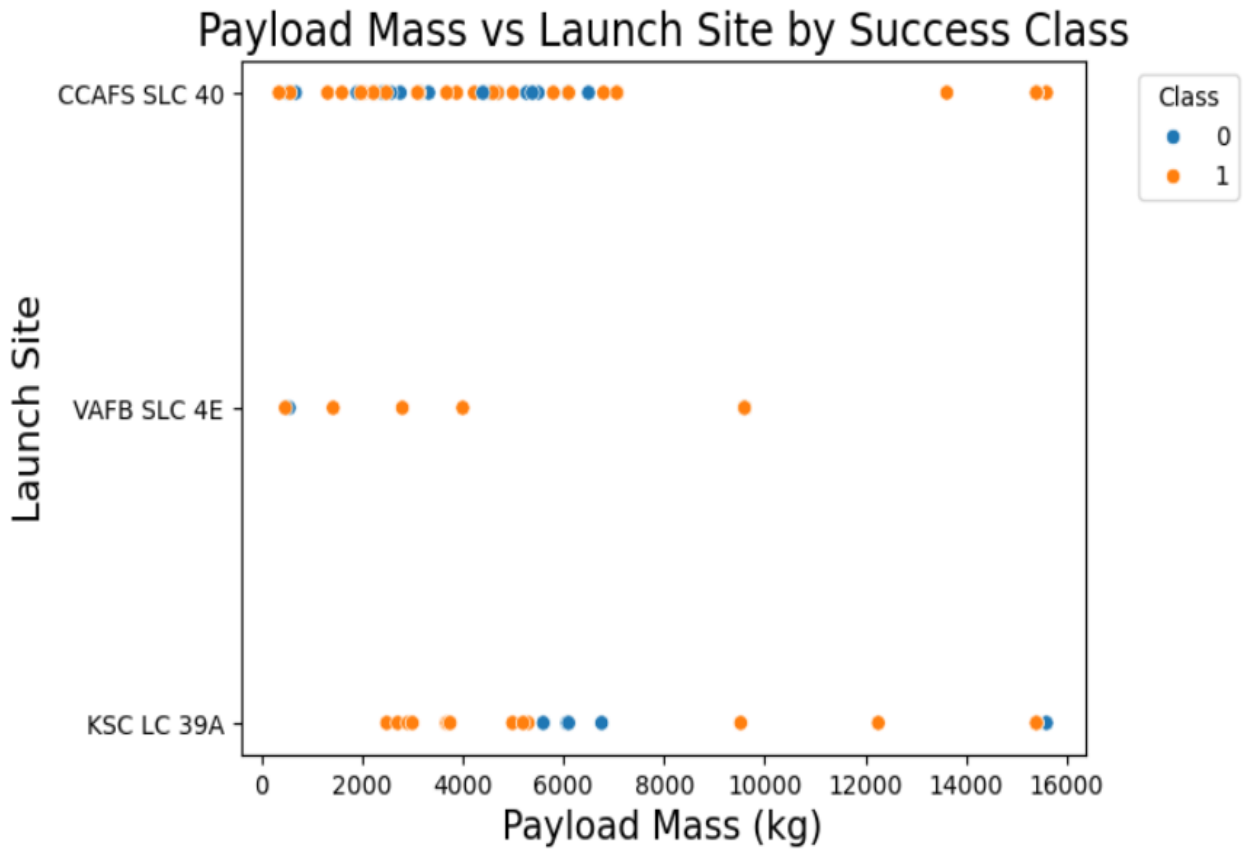
Predictive analysis methodology

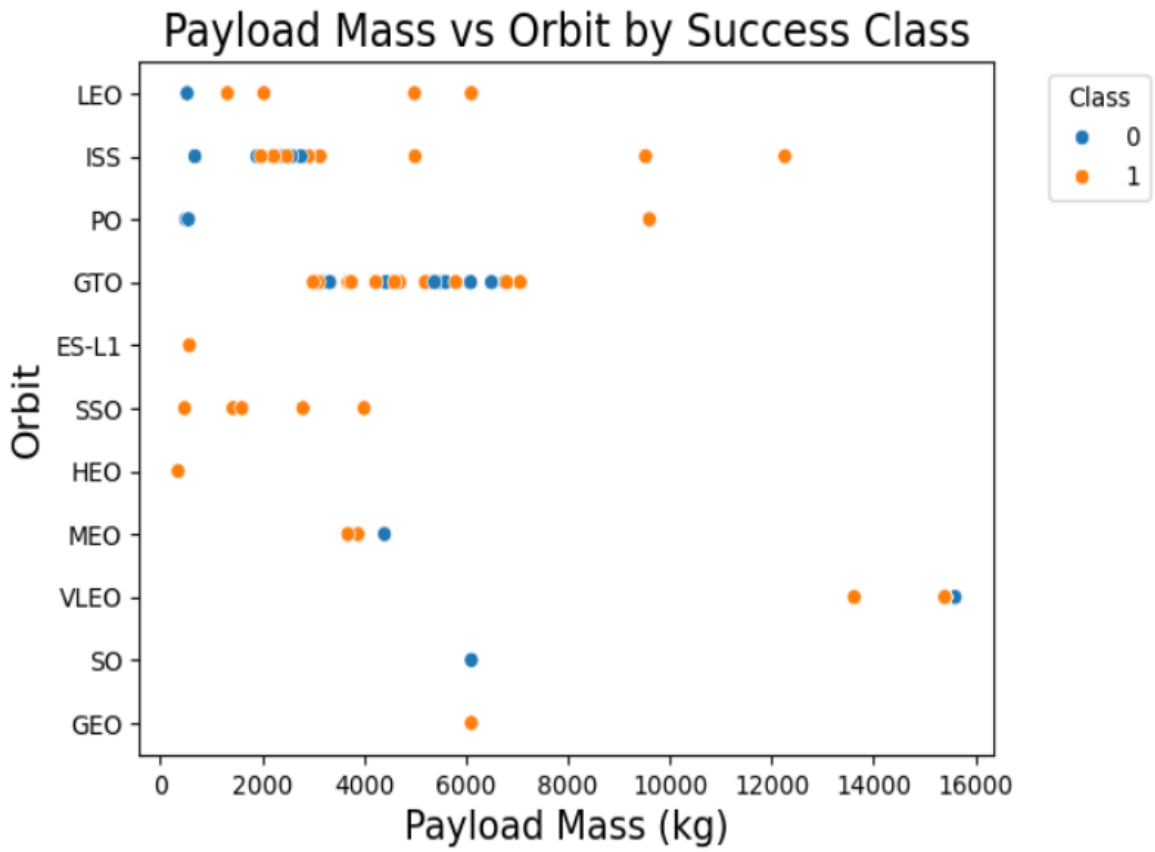
- **Objective**: To build and compare four supervised machine learning models to classify whether a mission's first stage will land successfully.
- **Algorithms Selected**:
  - Logistic Regression: Baseline model for binary classification.
  - Support Vector Machine (SVM): Effective in high-dimensional spaces.
  - Decision Tree: Non-linear model capable of capturing complex decision rules.
  - K-Nearest Neighbors (KNN): Instance-based learning for local pattern detection.
- Standardization: Applied StandardScaler to the feature set (X) to ensure all variables have a mean of 0 and standard deviation of 1, preventing features with larger scales (like Payload Mass) from biasing the models.
- Hyperparameter Tuning: Utilized GridSearchCV with 10-fold cross-validation for each model to identify the optimal parameters (e.g., regularization strength C, kernel types, and maximum tree depth).
- Data Splitting: Divided the processed dataset into Training (80%) and Testing (20%) sets to evaluate the models' generalization ability on unseen data.

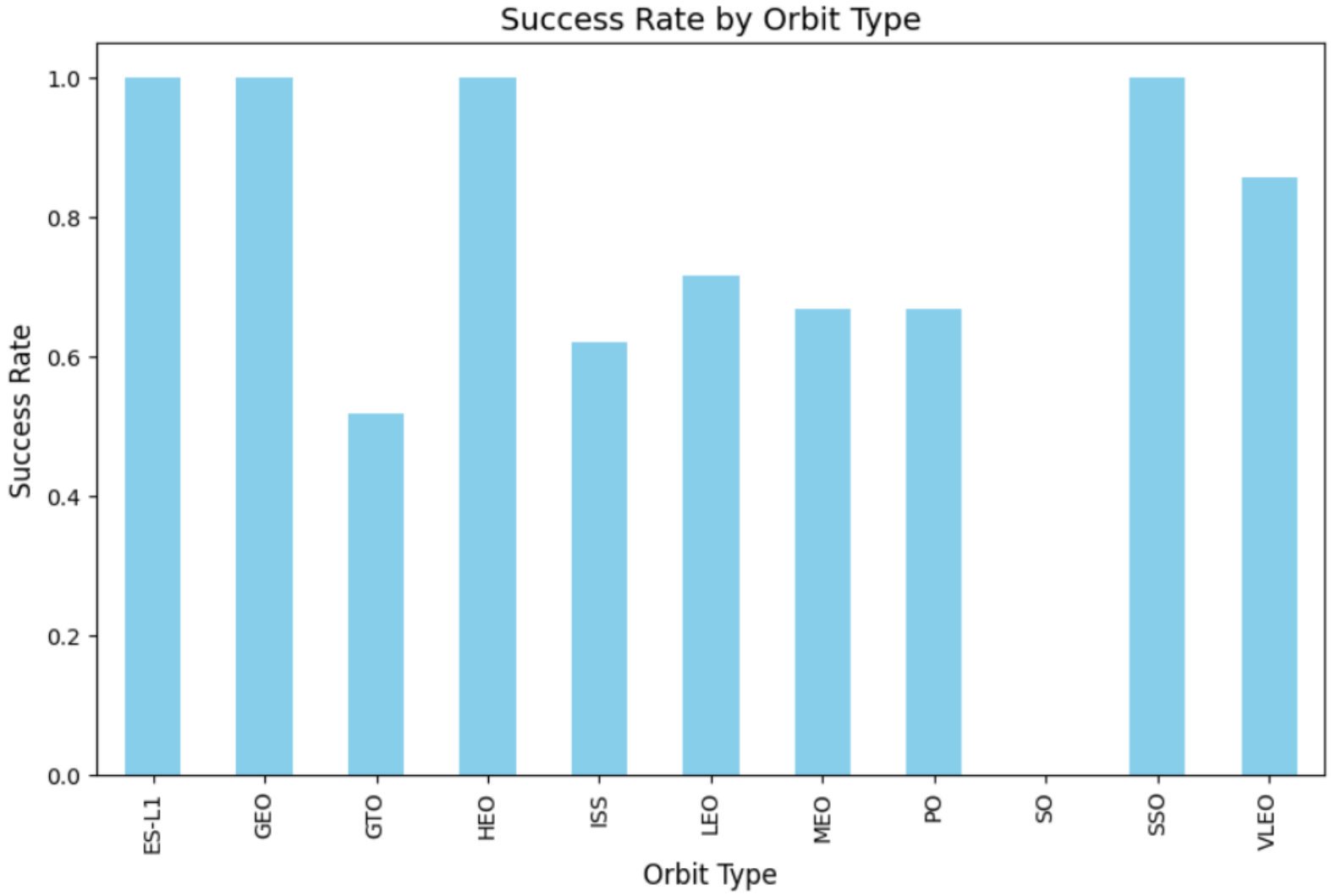
Key Findings:

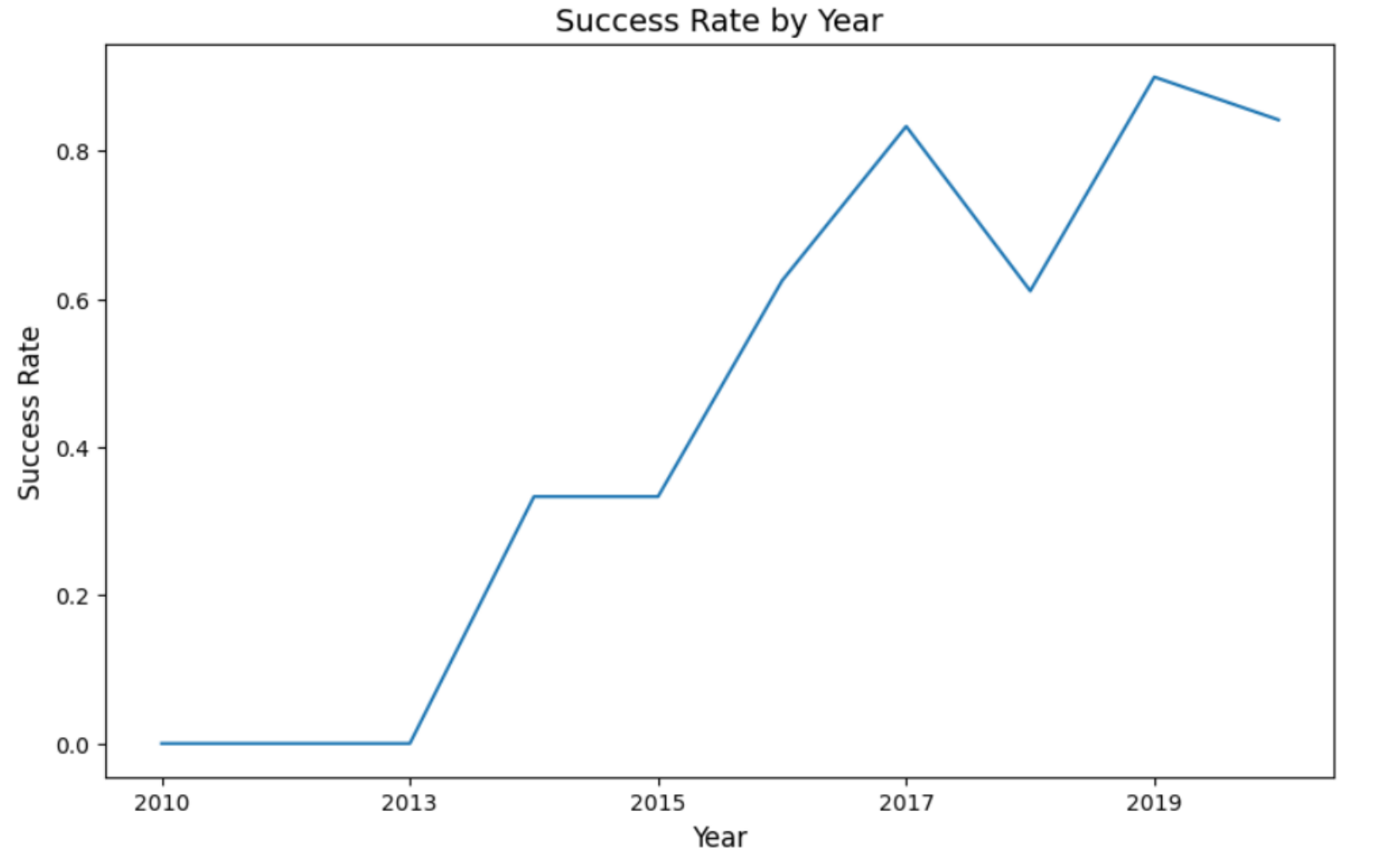
Top Performer: The **Decision Tree** achieved the highest accuracy (**88.89%**), correctly predicting more outcomes on the test set than its counterparts.

EDA with visualization results






Success Rate by Orbit Type
Success Rate
1.0
0.8
0.6
0.4
0.2
0.0
ES-L1
GEO
GTO
HEO
ISS
LEO
MEO
PO
SO
SSO
VLEO
Orbit Type


Success Rate by Year
Success Rate
0.0
0.2
0.4
0.6
0.8
2010
2013
2015
2017
2019
Year

EDA SQL Results

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "Payload_Mass__kg_" BETWEEN 4000 AND 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Task 10

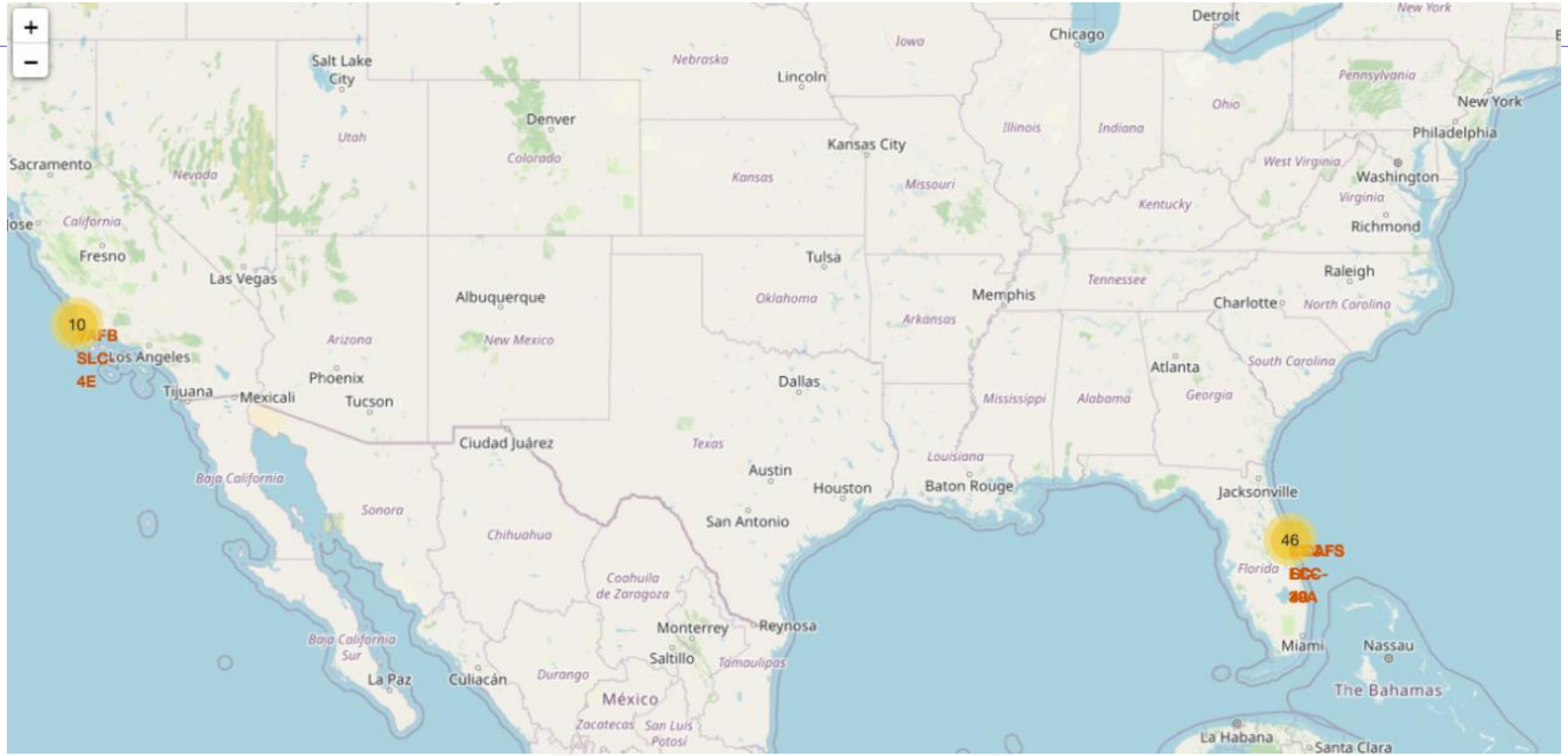
Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

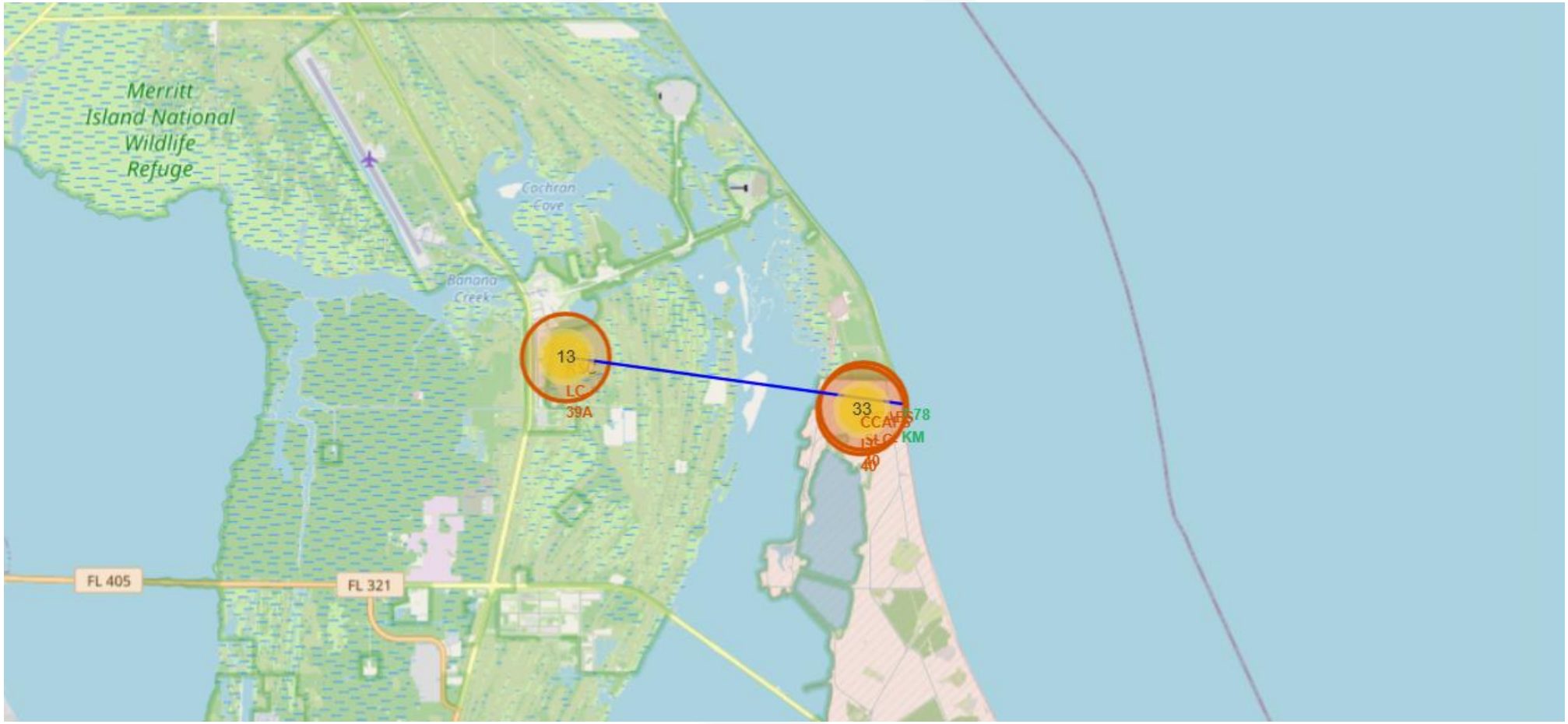
```
%sql SELECT Landing_Outcome, COUNT(*) as Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC
```

```
 * sqlite:///my_data1.db
Done.
```

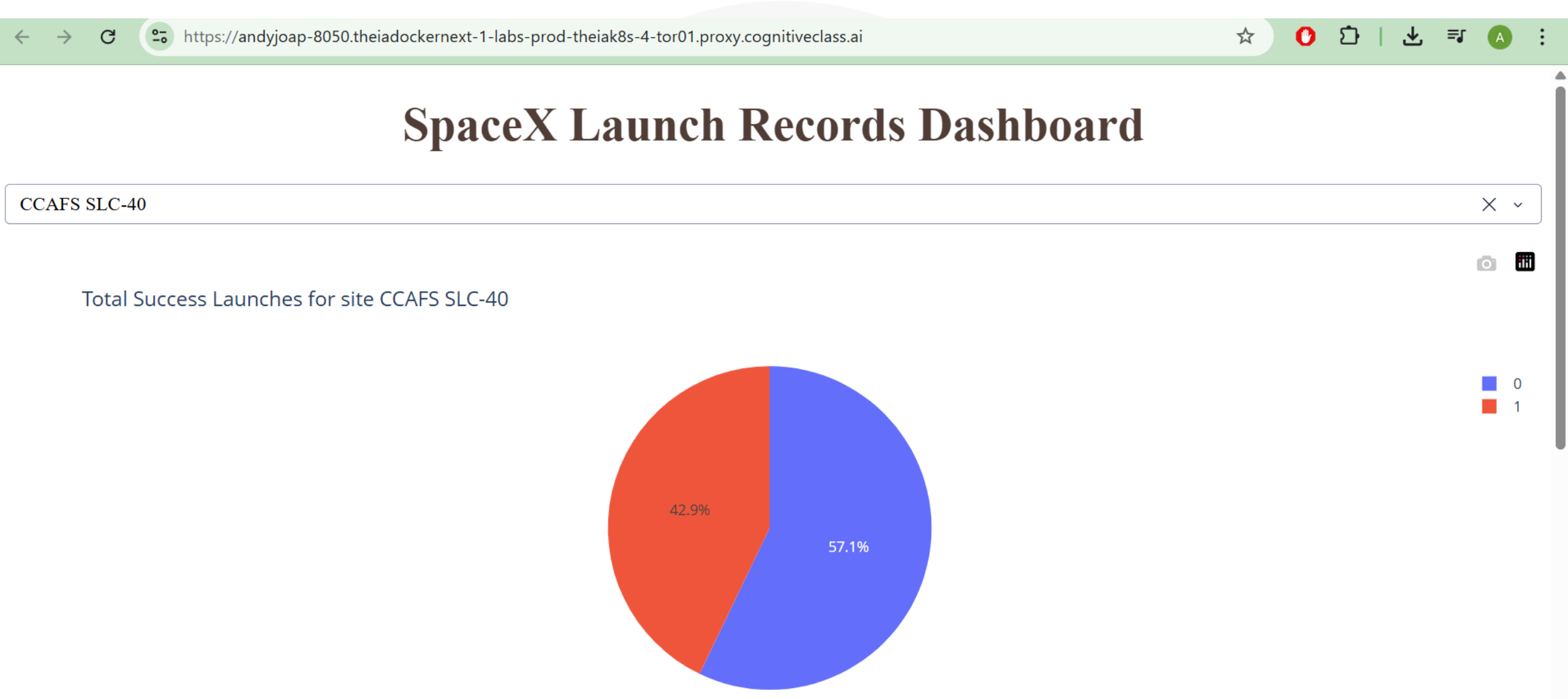
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Interactive map results using Folium


10
SLC
4E
Los Angeles
Tijuana
Mexicali
Fresno
Sacramento
Las Vegas
Salt Lake City
Denver
Phoenix
Tucson
Albuquerque
Ciudad Juárez
Lincoln
Kansas City
Tulsa
Dallas
Austin
Houston
San Antonio
Baton Rouge
Memphis
Chicago
Detroit
Atlanta
Charlotte
Raleigh
Richmond
Washington
Philadelphia
New York
Jacksonville
46
AFS
Miami
Nassau
The Bahamas
La Habana
Santa Clara
Monterrey
Reynosa
Saltillo
La Paz
Culiacán
México
California
Nevada
Utah
Colorado
Arizona
New Mexico
Nebraska
Kansas
Oklahoma
Texas
Iowa
Missouri
Arkansas
Louisiana
Mississippi
Alabama
Georgia
Florida
South Carolina
North Carolina
Tennessee
Kentucky
Virginia
West Virginia
Ohio
Indiana
Illinois
Pennsylvania
New York
Baja California
Sonora
Chihuahua
Coahuila de Zaragoza
Durango
Tamaulipas
Zacatecas
San Luis Potosí
Baja California Sur


Merritt
Island National
Wildlife
Refuge
Cochran
Cove
Banana
Creek
13
LC
39A
33
CCAFS
LC
40
KM
FL 405
FL 321

Plotly Dash dashboard results



https://andyjoap-8050.theiadockernext-1-labs-prod-theiak8s-4-tor01.proxy.cognitiveclass.ai

SpaceX Launch Records Dashboard

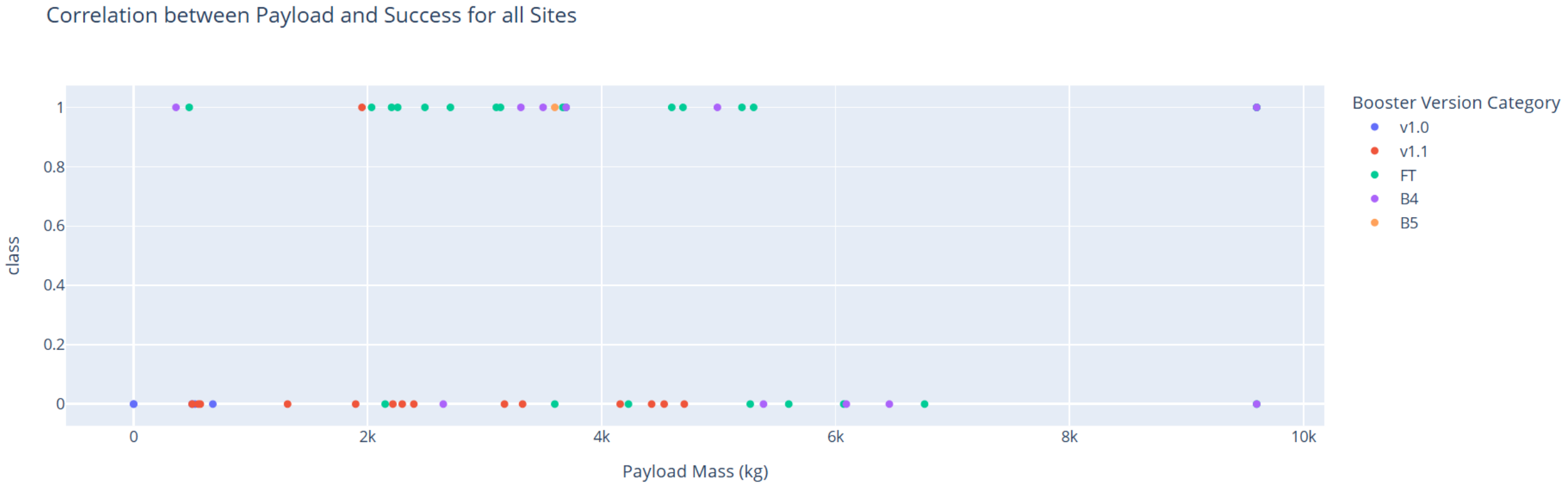
All Sites

Total Success Launches By Site

https://andyjoap-8050.theiadockernext-1-labs-prod-theiak8s-4-tor01.proxy.cognitiveclass.ai

Payload range (Kg):

0 — 0 — 2500 — 5000 — 7500 — 10000 — 10000



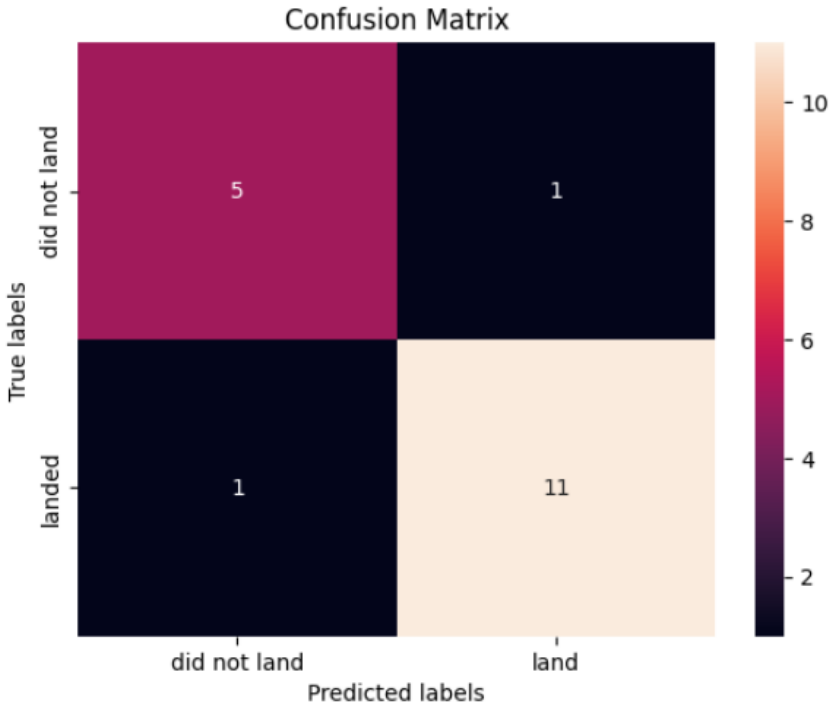
Predictive analysis (classification)

```
tree_score = tree_cv.score(X_test, Y_test)
print("Accuracy en Test:", tree_score)
```

```
Accuracy en Test: 0.8888888888888888
```

We can plot the confusion matrix

```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



- Decision Tree Score:

  0.88

CONCLUSION

- **Feasibility of Prediction**: The project successfully demonstrates that SpaceX Falcon 9 landing outcomes can be predicted with high accuracy (88.89%) using machine learning models.
- **Key Success Drivers**: Through our analysis, we identified that Payload Mass and Orbit Type are the most critical factors influencing the success of a first-stage recovery.
- **Strategic Insights**:Launch sites are strategically placed near the coast to ensure safety, confirming that geographical factors are as important as technical ones.The learning curve is evident: success rates have improved significantly as the number of flights increased.
- **Economic Impact**: By providing reliable landing forecasts, SpaceX can better manage its fleet of reusable rockets, potentially saving $40M+ per mission and optimizing global space logistics.
- **Future Work**: Further improvements could be achieved by incorporating real-time weather data (wind speeds, sea state) and increasing the dataset size as more launches occur.